Relations and orders exercises

1. On $A = \{1, 2, 3\}$, let $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$. Determine the five basic properties and the smallest equivalence relation containing R.
2. Represent $R = \{(a, b), (b, c), (c, a), (a, a)\}$ by a Boolean matrix. Compute entries of its square and identify missing transitive edges.
3. Order $S = \{2, 3, 4, 8, 12, 18\}$ by positive-integer divisibility. Find minimal and maximal elements, minimum and maximum if any, and draw the Hasse covers.
4. Prove that a partition's same-block relation is an equivalence relation whose classes are the original blocks.
5. Two DFAs begin with accepting/nonaccepting partitions. During refinement, states in one block have successors in one block for a particular input. Does this already justify merging? State the full stability condition.
6. Is the intersection $R = \preceq_1 \cap \preceq_2$ of two partial orders on one set always a partial order? Prove it or give a counterexample.
7. Derive the counts of reflexive, symmetric, antisymmetric, and asymmetric relations on n elements. Explain why symmetric and antisymmetric need not mean empty. Evaluate at n=3.
8. Apply Warshall to a→b→c→d, listing new pairs each round and final R^+,R^*. Use a directed three-cycle to explain why $R \cup R^2$ need not equal R^+.
9. For R={(1,2),(3,2)} on {1,2,3,4}, find its smallest equivalence closure and prove minimality. Is taking transitive closure and then adding only inverses and diagonal pairs sufficient?
10. In the poset of all positive divisors of 12, find bounds, glb, and lub for {2,3} and {4,6}. List every linear extension and use prefix words to prove completeness. What ceases to exist after restricting to {2,3,6,12}?
11. Prove that a unique maximal element of a finite nonempty poset is greatest, and give an infinite counterexample. Give a finite poset with top and bottom that is not a lattice, identifying a pair without a lub.
12. List every possible pair count of an equivalence relation on five elements and prove completeness using squared class sizes. Count partitions with exactly two blocks and explain why the blocks are not labeled.
13. For the text's complete six-state DFA, compute P_0,P_1,P_2,P_3 and the quotient table. Give a distinguishing suffix for each pair of its four classes and prove that every same-language DFA needs at least four states.

Relations and orders exercises (continued)

14. For the text's four-state Mealy table, compute all refinements and the quotient table; check output on 010. Add t with cells s/0 on both 0 and 1. Explain why t cannot merge with p and give a shortest witness.
15. For $a \sim b \iff 5 \mid (a-b)$ on $\mathbb{Z}$, decide whether $[a] \mapsto a^2 \in \mathbb{Z}$, $[a] \mapsto [a^2] \in \mathbb{Z}/5\mathbb{Z}$, and $[a] \mapsto (-1)^a$ are well-defined. Prove each answer.

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study

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